

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

First Semester of M. Sc. (Mathematics) Examination March 2018

MA715:Topology - I

March 28, 2018, Wednesday

Time 10.00 a.m. to 10.45 p.m.

Maximum marks 20

Multiple Choice Questions

Instructions:

1. Tick the correct answer only in this sheet.
2. You can use only non-programmable calculator. No other gadget is allowed in the examination hall.
3. Each question carries one mark.
4. In this question paper, $\mathbb{R}$ denotes the set of all real numbers with usual metric and $\mathbb{N}$ denotes the set of all natural numbers unless specified. Other notations are standard.

Multiple Choice Questions

Q-I

Choose the correct answer from the given options in the following:

[20]

- (1) Which of the following subset of $\mathbb{R}$ is uncountable?
 - (a) The set of rational numbers.
 - (b) $[0, 1]$
 - (c) $\{m + n\sqrt{3} : m, n \in \mathbb{N}\}$
 - (d) $\{a \in [0, 1] : a \text{ is rational}\}$
- (2) There is no one to one function from X on to its proper subset. Then _____.
 - (a) X is finite
 - (b) X is infinite countable
 - (c) X is uncountable
 - (d) None of these
- (3) Define a metric on $\mathbb{R}$ by $d(x, y) = \min\{1, |x - y|\}$, $x, y \in \mathbb{R}$. Then the open sphere $S_3(0)$ = _____.
 - (a) $(0, 3)$
 - (b) $(-3, 3)$
 - (c) $\mathbb{R}$
 - (d) $\{0\}$
- (4) If $x_n = ______$ for each $n \in \mathbb{N}$, then the sequences $\{x_n\}$ is convergent?
 - (a) $(-3)^n$
 - (b) 3^{-n}
 - (c) $n/3$
 - (d) $n^{2/3}$
- (5) Which of the following is a bijective function from $\mathbb{R}$ to $\mathbb{R}$?
 - (a) $f(x) = x + 2$
 - (b) $f(x) = x^2$
 - (c) $f(x) = e^x$
 - (d) $f(x) = \sin(x)$
- (6) Let (X, d) be a metric space and $A \subset X$. Then A is open if _____.
 - (a) $\partial A \subset A$
 - (b) $A \subset \overline{A}$
 - (c) $\text{int}(A) \neq \emptyset$
 - (d) $\text{int}(A) = A$
- (7) Let $X \neq \emptyset$. Define $d(x, y) = \begin{cases} 1 & \text{if } x = y \\ 0 & \text{if } x \neq y \end{cases}$ be a metric on X . Then _____.
 - (a) a singleton set in X is not open
 - (b) each subset of X is open
 - (c) X has no open subset
 - (d) X has no closed subset
- (8) The set of all limit points of the set $(0, 1)$ in the usual metric of $\mathbb{R}$ is _____.
 - (a) $(0, 1)$
 - (b) $\{0, 1\}$
 - (c) $[0, 1]$
 - (d) $\emptyset$
- (9) Let (X, d) be a metric space and $A \subset X$. Which of the following is **false**?
 - (a) $\overline{A}$ is a closed subset of X .
 - (b) A has no limit point then A is closed
 - (c) $\text{Int}(A)$ is an open subset of X .
 - (d) If $\text{Int}(A) \neq \emptyset$, then A is closed.
- (10) Let (X, d) be a metric space and $A \subset X$. Then _____.
 - (a) $\partial(A)$ is close
 - (b) $\partial(A) \subset \text{int}(A)$
 - (c) $\partial(A)$ is open
 - (d) $\partial(A) \subset A^c$
- (11) Which of the following subsets of $\mathbb{R}$ is closed?
 - (a) $\mathbb{N}$
 - (b) $(0, 1)$
 - (c) rational numbers
 - (d) irrational numbers
- (12) Which of the following is a dense subsets of $\mathbb{R}$?
 - (a) $\mathbb{N}$
 - (b) $\mathbb{Z}$
 - (c) rational numbers
 - (d) $(-\infty, a) \cup (b, \infty)$, $a < b$
- (13) Let (X, d) be a metric space and $a \in X$. Let $A = \{a\}^c$. Then _____.
 - (a) A is open
 - (b) A is nowhere dense in X
 - (c) A is dense in X
 - (d) A is closed
- (14) Which of the following subsets of $\mathbb{R}$ are compact?
 - (a) $\mathbb{Z}$
 - (b) $(0, \infty)$
 - (c) $[0, \infty)$
 - (d) $\{1, 2, 3, \dots, n\}$
- (15) Let $X = \mathbb{N}$ with discrete metric space. Which of the following is an open cover of X ?
 - (a) $\{\{2n\} : n \in \mathbb{N}\}$
 - (b) $\{\{n\} : n \in \mathbb{N}\}$
 - (c) $\{\{2n + 1\} : n \in \mathbb{N}\}$
 - (d) $\{\{n^2\} : n \in \mathbb{N}\}$

- (16) A closed and bounded subset of a metric space is _____.
 (a) compact (b) complete (c) open (d) none of these
- (17) $f(x) = 1/x$ is not uniformly continuous on _____.
 (a) $[1, 2]$ (b) $(0, 2)$ (c) $[2, 4)$ (d) $(2, 4)$
- (18) A compact subset of metric space is _____.
 (a) Countable (b) finite (c) closed (d) open
- (19) Let X be a metric space and $A, B \subset X$ with A is nowhere dense. Then _____ is nowhere dense..
 (a) $A \cup B$ (b) $A^c \cup B$ (c) $A \cup B^c$ (d) $A \cap B$
- (20) Let $X = \mathbb{N}$ with discrete metric space. Then _____.
 (a) X is compact (d) X is complete
 (b) There is a subset of X which is not open (c) none of these

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

First Semester of M. Sc. (Mathematics) Examination December 2017

MA715 : Topology - I

March 28, 2018, Wednesday

Time 10.45 a.m. to 1.00 p.m.

Maximum marks 50

Instructions:

1. Section I and Section II must be written in separate answer books.
2. You can use only non-programmable calculator. No other gadget is allowed in the examination hall.
3. Figures to the right indicate marks.
4. In this question paper, $\mathbb{R}$ denotes the set of all real numbers with usual metric unless specified. Other notations are standard.

SECTION - I

- Q-II (A) Attempt any TWO of the following: [8]
- (1) Prove that the set $[0, 1]$ is uncountable.
 - (2) Let (X, d) be a metric space. Define $\bar{d}(x, y) = \min\{1, d(x, y)\}$, $x, y \in X$. Show that the metrics d and $\bar{d}$ generate same topology.
 - (3) Let (X, d) be a metric space. Show that arbitrary union of open sets in X is open in X .
- (B) Attempt any SIX of the following: [12]
- (1) Show that the set of all rational numbers is countable.
 - (2) Let (X, d) be a metric space and $Y \subset X$ be complete. Prove that Y is closed.
 - (3) Let X be a nonempty set. If X is compact with discrete metric. Show that X is finite.
 - (4) Let (X, d) be a metric space. Let $r > 0$ and $x \in X$. Define the open sphere $S_r(x)$. Show that $S_r(x)$ is open in X .
 - (5) Let (X, d) be a metric space; A and B be subsets of X . Show that $\text{Int}(A \cap B) = \text{Int}(A) \cap \text{Int}(B)$.
 - (6) Let (X, d) be a metric space and $A \subset X$. Show that A is open in X iff A^c is closed in X .
 - (7) Let (X, d) be a metric space and $A \subset X$. Show that $\text{int}(A)$ is an open subset of A .
 - (8) Let (X, d) be a metric space and $A \subset X$. Define boundary of A . Show that $\partial A = \overline{A} \cap \overline{A^c}$.
 - (9) Show that a compact subset of a metric space is closed.

SECTION - II

- Q-III (A) Attempt any THREE of the following: [18]
- (1) Let X be a complete metric space and $\{F_n\}$ be a sequence of closed subsets of X such that $F_n \subset F_{n+1}$ for each n and $\text{dia}(F_n) \rightarrow 0$. Show that $\bigcap \{F_n : n = 1, 2, 3, \dots\}$ is a singleton.
 - (2) State and prove Holder's inequality.
 - (3) Let X be a compact metric space. Show that X has Bolzano-Weierstrass property.
 - (4) Let X be a sequentially compact metric space. Show that X is compact.
 - (5) Let X be a compact metric space and Y be a metric space. If $f: X \rightarrow Y$ is continuous, show that f is uniformly continuous.

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- (B) Attempt any THREE of the following: [12]
- (1) Let X be a metric space and $A \subset X$. Show that A is nowhere dense iff A^c contains a dense open set.
 - (2) Let (X, d_1) and (Y, d_2) be metric spaces and $f: X \rightarrow Y$ be continuous. Show that whenever $\{x_n\}$ converges to x in X , then $\{f(x_n)\}$ converges to $f(x)$ in Y .
 - (3) Let (X, d) be a metric space and $A \subset X$. Show that $\overline{A} = \{x \in X : d(x, A) = 0\}$
 - (4) Show that every convergent sequence is Cauchy. Also show that every Cauchy sequence is bounded.
 - (5) Show that closed subset of a compact metric space is compact
